

Amit Kumar Singh

National PhD in Earth Observation | Sapienza University of Rome, Italy | University of Naples Federico II, Italy
Visiting PhD Scholar | Cornell University, NY, USA

Contacts: amitkumar.singh@uniroma1.it | (+39) 3518527097 | <https://www.linkedin.com/in/amit896689/>

EDUCATION

Doctor of Philosophy | *National PhD in Earth Observation, 39th Cycle* **Nov. 2023 – Oct. 2026**
Department of Civil, Building and Environmental Engineering | Sapienza University of Rome, Italy

- **Tentative Thesis Title:** *Development of Machine-Learning Framework for Providing Land Use/Land Cover Thematic Products Using Earth Observation Copernicus Data.*
- **Related Work:** *Classification of Irrigated Cropland Area in Southern Italy Region using AI/ML Algorithms*

Master of Technology | *Civil Engineering (Hill Area Development Engineering)* | 66 Credits **2018 – 2020**
Madan Mohan Malaviya University of Technology, Gorakhpur, India

- **CGPA: 8.79/10** | First Division – Honours | Gold Medalist – **Rank: 1st**
- **Thesis Title:** *Assessment of Long-Term Status (2000 – 2015) of Snow Cover Area in North Western Himalayan Region Using Remote Sensing.*

Bachelor of Technology | *Civil Engineering* | 201 Credits **2013 – 2017**
Dr. A.P.J. Abdul Kalam Technical University, Lucknow (U.P.) India – *formerly known as U.P.T.U*

- **Result: 79.88%** | First Division – Honours
- **Project Title:** *Design of Highways with Shifting Lane Dividers.*

RESEARCH EXPERIENCE

Visiting PhD Research Scholar | Cornell University, USA **Apr. 2026 – Sep. 2026**
School of Civil and Environmental Engineering, Cornell Duffield College of Engineering, Ithaca, NY, United States

- **Job Responsibilities:** *Develop crop mapping models using Earth observation data to support irrigation water and energy optimization in critical infrastructure systems.*

Project Associate | Indian Institute of Technology Roorkee, India **Apr. 2023 – Oct. 2023**
Department of Computer Science & Engineering, IIT Roorkee, Roorkee, Uttarakhand, India

- **Project Title:** *Landslide Monitoring, Forecasting and Mitigation System – funded by ISRO*
- **Job Responsibilities:** *Development of a landslide monitoring system which can spontaneously monitor and report the occurrence of landslide and land flow along with its extent.*

Junior Research Fellow (JRF) | Indian Institute of Technology Delhi, India **Jan. 2023 – Mar. 2023**
Civil Engineering Department, IIT Delhi, New Delhi, India

- **Project Title:** *Estimation of freshwater discharge at river mouth to Ocean through Altimetry and Hydrological modelling techniques (SAMUDRA) – funded by ISRO.*
- **Job Responsibilities:** *Hydrological modelling, in situ discharge and bathymetric measurements, satellite remote sensing.*

Research Intern | Indian Institute of Technology Bombay, India **Nov. 2022 – Dec. 2022**
Civil Engineering Department, IIT Bombay, Mumbai, Maharashtra, India

- **Project Title:** *Durability of construction materials – funded by SERB.*
- **Job Responsibilities:** *Characterisation of construction materials, laboratory testing with the preparation of manuscript and research report to publish conference paper.*

Teaching Assistant (JRF) | National Institute of Technology Agartala, India **Feb. 2021 – Dec. 2022**
Civil Engineering Department, NIT Agartala, Tripura, India

- **Project Title:** *Impact assessment of environmental parameters on landslide risk using geospatial analysis.*
- **Job Responsibilities:** *To guide UG & PG scholars in their final year project and research paper writing.*
- **Mentored Projects:** i) *Hazard mapping and risk assessment modelling of ONGC Tripura Power Company using ALOHA and Geo-spatial tools.*
ii) *Development of WebGIS application for payment system touchpoints in NIT Agartala.*

PUBLICATIONS ([Google Scholar](#))

1. **Singh, A. K.**, & Madhubala. (2026). High-Resolution Global Methane Mapping: Advances in Satellite Remote Sensing, Machine Learning, and Policy Frameworks. *Methane*, 5(3), 21. <https://doi.org/10.3390/methane5030021>
2. **Singh, A. K.**, Belfiore, O. R., Pugliano, G., & D'Urso, G. (2024, December). Rice Crop Growth Stage Monitoring Using Harmonized Sentinel-2 and SAR Data in Google Earth Engine. In *2024 International Conference on Emerging Technologies and Innovation for Sustainability (EmergIN)* (pp. 576-579). IEEE.
3. **Singh, A. K.**, Belfiore, O. R., Pugliano, G., & D'Urso, G. (2024, November). Flood Hazard Assessment and Cropland Damage Estimation Using Harmonized Sentinel-2 and SAR Data. In *2024 International Conference on IoT, Communication and Automation Technology (ICICAT)* (pp. 859-863). IEEE.
4. **Singh, A. K.**, Pandey, D. K., Mishra, P., & Ghosh, S. (2024, December). Risk Assessment of Natural Gas Pipeline Dispersion Using ALOHA Modelling and Remote Sensing Techniques. In *2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS)* (pp. 1-4). IEEE.
5. **Singh, A. K.**, Pujara, K., Pugliano, G., Belfiore, O. R., & D'Urso, G. (2025, August). A Hybrid Deep Learning and Object Based Image Analysis Framework for Enhanced Landslide Detection. In *IGARSS 2025-2025 IEEE International Geoscience and Remote Sensing Symposium* (pp. 1328-1331). IEEE.

SKILLS

Geospatial & Bathymetric Analysis
ArcGIS, QGIS, ADCP, DGPS, SNAP, ERDAS Imagine, AgiSoft Metashape

Programming Languages
Python and R Programming
AI / ML and Google Earth Engine

Material Testing & Designing
SEM, XRD, UPV, AutoCAD, STAAD.Pro

TRAINING & WORKSHOPS

Workshop on Climate Modelling | *IISc Bangalore, India*
Workshop on Flood Modelling | *MMMUT, Gorakhpur, India*
Capacity Building in Geo-Spatial Technologies | *NIT Agartala, India*
Multi-Hazard Risk Management Solution for Cities, Industries and Business | *IIT Indore, India*
GATHERS Advanced School 2024 | *Sapienza University, Italy*
Advanced Geospatial Techniques for Coastal and Marine Applications | *IIT Madras, India*
EPOS Summer School 2025 | *Olot, Spain*
IEEE IGARSS 2025 - Australian Challenges Summer School | *Griffith University, Australia*
Satellite Based Hydrological Data Assimilation | *Aalborg University, Denmark*
14th Advanced Training Course on Land Remote Sensing: Agriculture | *Aristotle University of Thessaloniki, Greece*
2025 IEEE GRSS Microwave Monitoring of Water with Applications (MAMI WATA) | *Sapienza University, Italy*

ACHIEVEMENTS

Certifications:

Disaster Preparedness | *University of Pittsburgh*
Energy Within Environmental Constraints | *Harvard University*
Machine Learning | *Stanford University*
Excel Crash Course | *Corporate Finance Institute® (CFI)*
Computer Programming for Everyone | *University of Leeds*
Programming for Everybody | *University of Michigan*

Awards:

ISSNAF Fellowship – Cornell University, USA
IEEE GRSS Travel Grant – IGARSS 2025, Australia
IEEE GRSS (IGARSS 2025) Data Challenge | Winner
Inter Branch Badminton Tournament | Winner
Civil Engineering Branch Logo Designing | Winner
A.C.C. Mastermind Quiz | National Level Winner

REFERENCES

Dr. Stefano Galelli**(Professor)**

School of Civil & Environmental Engineering,
Cornell University, NY, United States.

Email: galelli@cornell.edu

Mobile: +1 (607) 255 2351

Dr. Guido D'urso**(Professor)**

Department of Agriculture,
University of Naples Federico II, Italy.

Email: durso@unina.it

Mobile: (+39) 081-2539418

Dr. Giovanni Pugliano**(Professor)**

DICEA,
University of Naples Federico II, Italy.

Email: giovanni.pugliano@unina.it

Mobile: (+39) 081-7683374